A. Synthetic Quadratic Optimization Problem

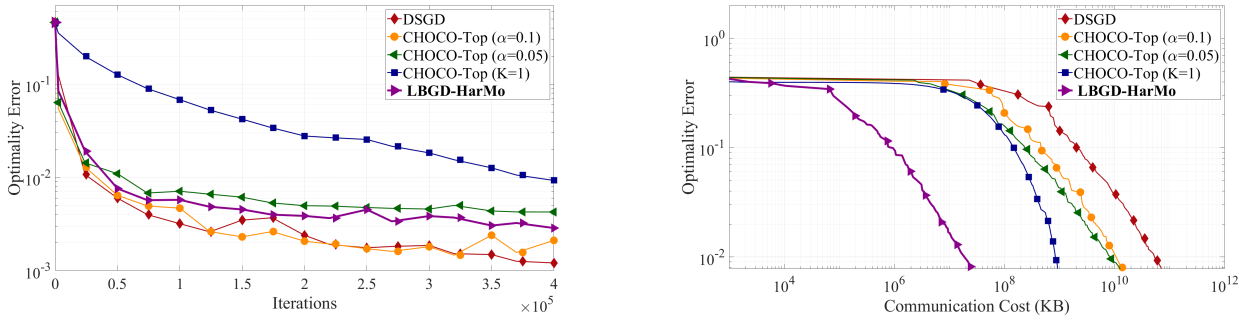
Table 1. Total number of iterations and communication cost under different experimental settings on the synthetic quadratic optimization problem. The table reports sensitivity results with respect to the number of clients, network topology, problem dimension, harmonic sequence choice, number of asynchronous clients, communication frequency, quantization precision, and the compression ratio α for CHOCO and MoTEF.

Number of clients	**Iteration**	**Comm. cost (MB)**
9	39,472	1.36
25	25,810	2.46
64	28,020	6.84
100	36,713	14.00
Topology	**Iteration**	**Comm. cost (MB)**
Fully-connected	9,550	10.93
Torus	28,523	5.44
Fixed ER	38,233	4.38
Ring	25,810	2.46
Time-varying ER	79,554	9.10
Dimension	**Iteration**	**Comm. cost (MB)**
4	17,056	1.67
8	25,810	2.46
16	31,354	2.99
100	33,105	3.16
Sequence	**Iteration**	**Comm. cost (MB)**
ψ_{HarMo}	25,810	2.46
$\psi_{\boldsymbol{e}_i}(t)$	24,224	2.31
$\psi_{\mathrm{Gaussian}}(t)$	25,711	2.45
Number of asynchronous clients	**Iteration**	**Comm. cost (MB)**
5	28,879	2.48
10	29,985	2.29
15	37,063	2.47
20	41,027	2.35
Comm. frequency	**Iteration**	**Comm. cost (MB)**
2	29,985	2.29
4	42,545	2.84
6	56,287	3.58
8	69,185	4.29
Quantization precision (bit)	**Iteration**	**Comm. cost (MB)**
3	47,653	4.54
4	38,717	3.69
8	25,810	2.46
16	24,789	4.73
Compression ratio α (CHOCO)	**Iteration**	**Comm. cost (MB)**
0.125	38,074	29.05
0.375	37,118	84.96
0.625	33,939	129.47
0.875	33,546	179.15
Compression ratio α (MoTEF)	**Iteration**	**Comm. cost (MB)**
0.125	4,180	3.19
0.375	3,812	8.72
0.625	3,775	14.40
0.875	3,687	19.69

B. Logistic Regression with Strongly Convex Regularizer

Table 2. Per-client communication cost per iteration, test accuracy, and runtime for different algorithms on the logistic regression task. Experiments are conducted on $n = 9$ IID clients over a ring topology. The table reports the uncompressed baseline (DSGD), CHOCO with Top-α compression ($\alpha = 0.1, 0.05$) and the extreme Top-1 setting ($K = 1$), MoTEF with Top-α compression ($\alpha = 0.1$), LCC with Sign compression (1 bit), and HarMo ($m = 16$).

METHOD	ALGORITHM	COMM. COST (KB)	ACCURACY (%)	RUNTIME (S)
NONE	DSGD	31.25	88.44	454
TOP ($\alpha = 0.1$)	CHOCO	6.25	88.23	957
TOP ($\alpha = 0.05$)	CHOCO	3.125	87.56	951
TOP ($K = 1$)	CHOCO	0.031	86.91	952
TOP ($\alpha = 0.10$)	MOTEF	6.25	87.42	1497
SIGN	LCC	0.99	86.82	609
HARMO ($m = 16$)	**LBGD**	**0.008**	**87.84**	**734**



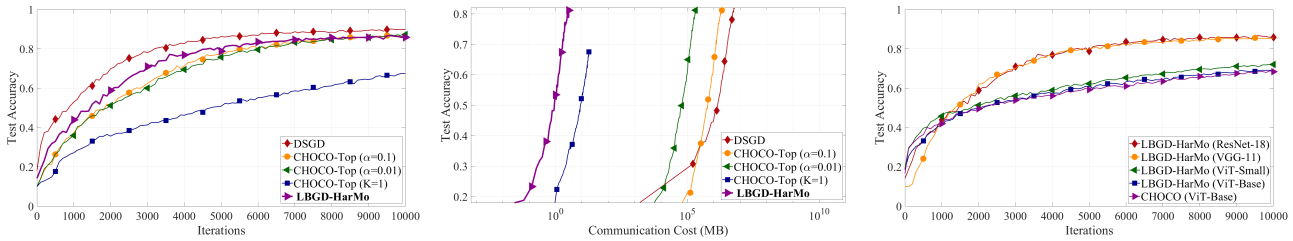
(a) Convergence rate *(b)* Communication cost

Figure 1. Comparison of DSGD, CHOCO with Top-α compression ($\alpha = 0.1, 0.05$), CHOCO with the extreme Top-1 setting ($K = 1$), and LBGD-HarMo ($m = 16$) on the logistic regression task with $n = 25$ clients over a ring topology. (a) shows the convergence behavior in terms of optimality error versus iterations, and (b) shows the trade-off between optimality error and communication cost.

C. Neural Network Training

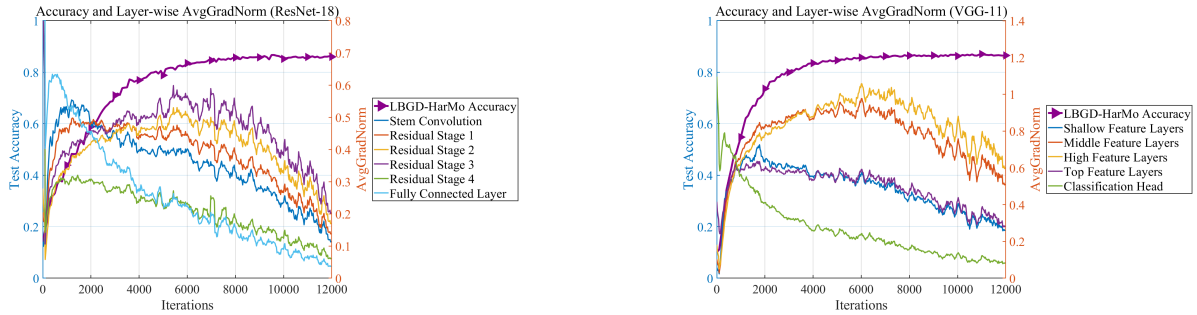
Table 3. Per-client communication cost per iteration, training accuracy, and runtime for different algorithms in neural-network training. Experiments are conducted on $n = 9$ IID clients over a ring topology. The table reports uncompressed baselines (FedAvg and DSGD), CHOCO with Top-α compression ($\alpha = 0.1, 0.01$) and the extreme Top-1 setting ($K = 1$), LCC with Sign compression (1 bit), and HarMo ($m = 24$). Results are shown on ResNet-18, and HarMo is further evaluated on VGG-11, ViT-Small, and ViT-Base; CHOCO on ViT-Base is also included for comparison. The relatively weaker performance of ViT models is likely due to their higher data demand and weaker built-in inductive bias on CIFAR-10, which make optimization under decentralized compressed training more challenging.

METHOD	ALGORITHM	MODEL	COMM. COST (MB)	ACCURACY (%)	RUNTIME (MIN)
NONE	FEDAVG	RESNET-18	805	89.92	99
NONE	DSGD	RESNET-18	179	90.09	84
TOP ($\alpha = 0.1$)	CHOCO	RESNET-18	36	87.03	209
TOP ($\alpha = 0.01$)	CHOCO	RESNET-18	4	86.38	213
TOP ($K = 1$)	CHOCO	RESNET-18	0.0003	67.47	207
SIGN	LCC	RESNET-18	5	81.94	147
HARMO ($m = 24$)	**LBGD**	**RESNET-18**	**0.0001**	**86.69**	**203**
HARMO ($m = 24$)	LBGD	VGG-11	0.001	86.49	185
HARMO ($m = 24$)	LBGD	VIT-SMALL	0.0002	72.06	372
HARMO ($m = 24$)	LBGD	VIT-BASE	0.0007	69.12	624
TOP ($\alpha = 0.1$)	CHOCO	VIT-BASE	265	68.38	675



(a) Convergence rate *(b)* Communication cost *(c)* Different Models

Figure 2. Comparison of DSGD, CHOCO with Top-α compression ($\alpha = 0.1, 0.01$), CHOCO with the extreme Top-1 setting ($K = 1$), and LBGD-HarMo ($m = 24$) for neural-network training on $n = 9$ IID clients over a ring topology. (a) shows the test accuracy versus iterations, and (b) shows the test accuracy versus communication cost. (c) compares the convergence behavior of LBGD-HarMo across different model architectures, with CHOCO on ViT-Base included as a reference. DSGD achieves the highest final accuracy, while LBGD-HarMo attains competitive performance with substantially lower communication cost and exhibits a more favorable communication–accuracy trade-off than CHOCO under aggressive compression. Across different models, HarMo remains effective, although the ViT-based models show weaker accuracy than the CNN-based models (ResNet-18 and VGG-11) in this setting.



(a) Layer-wise update trajectories (ResNet-18) *(b)* Layer-wise update trajectories (VGG-11)

Figure 3. Layer-wise update trajectories of LBGD-HarMo for different network architectures. The left vertical axis denotes the test accuracy, shown by the purple curve, while the right vertical axis denotes the layer-wise average gradient norm (AvgGradNorm), shown by the remaining colored curves. For ResNet-18, layers are grouped into the stem, residual stages, and final fully connected layer. For VGG-11, layers are grouped into feature layers of different depths and the classification head. Different layer groups exhibit distinct update patterns during training.

D. Comparison of Compression Methods

Table 4. Comparison of communication cost, computational complexity, and storage complexity for different compression methods, including Top-α, Sign, Sketch, Low-Rank, and HarMo. Here, d denotes the update dimension, α is the retained ratio in Top-α, b is the sketch size, r is the target rank in Low-Rank compression, p and n denote the two dimensions of the factorized matrix, and m is the number of transmitted bits in HarMo.

METHOD	COMMUNICATION COST	COMPUTATIONAL COMPLEXITY	STORAGE COMPLEXITY
TOP-α	$64\lceil \alpha d \rceil$	$\mathcal{O}(d \log d)$	$\mathcal{O}(d)$
SIGN	$d + 32$	$\mathcal{O}(d)$	$\mathcal{O}(d)$
SKETCH	$32b$	$\mathcal{O}(bd)$	$\mathcal{O}(bd)$
LOW-RANK	$32r(p+n)$	$\mathcal{O}(rd)$	$\mathcal{O}(r(p+n))$
HARMO	m	$\mathcal{O}(d)$	$\mathcal{O}(d)$